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EFFECT OF WEAK CONVECTION ON LAMELLAR SPACING OF EUTECTICS

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Abstract—An asymptotic expansion method is proposed to solve the convection–diffusion equation which governs the directional solidification of eutectic alloys under the condition of weak convection at low Peclet number. The so-called weak convection is defined here as the condition for which the velocity of convective flow and its gradient ahead of the S/L interface are both relatively small. Our analysis shows that such weak convective flow will give rise to an increase of eutectic lamellar spacing since the convection produces a slight shift of solute concentration profile at the interface and a decrease of eutectic growth undercooling. Our analytical results are compared with other models and have been applied to some practical cases such as weak forced convection induced by a rotating disk and natural convection induced by solutal and thermal differences in solidification processing. Calculations for two typical eutectic alloys, Pb–Sn and Al–Cu, are performed in order to reveal schematically the effect of weak convective flow on eutectic growth.
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1. INTRODUCTION

The effect of convection on eutectic growth has been a promising subject of many theoretical and experimental studies in the past few decades [1–7]. In the early 1970s, it was suggested that convective flow produces little effect on eutectic growth except by inducing a perturbation of growth temperature during the solidification process [1, 2]. This consideration arose from the fact that some researchers presumed a "stagnant film" [2] in the solute diffusion boundary layer just in front of the S/L interface, which has a thickness much larger than the eutectic interphase spacing. However, in the 1980s some theoretical analyses and experimental studies showed that convective flow resulted in a change of eutectic spacing during directional solidification [3–7]. Quenisset and Naslain [3] firstly obtained a theoretical solution for the effect of convection on the eutectic spacing as follows:

$$\lambda^2 = \frac{A}{1 - BG_u\lambda^2/D}$$

where λ is eutectic spacing under the existence of convection, V is growth velocity, D is solute diffusion coefficient, G_u is the gradient of flow velocity and A, B are constants. The following assumptions in their analysis, however, limited the application of their theory: (i) there is a presumed characteristic diffusion distance in front of the S/L interface, $\Phi/2$, where $\Phi/2 = (\lambda/2)(1 - \zeta\mathrm{Pe})$, ζ being the volume ratio of β phase to α phase and Pe $(= U\zeta/D)$ the flow *Peclet* number with U $(= G_u\lambda/8)$ the characteristic velocity of fluid flow; (ii) the considered eutectic is an ideal system where the eutectic composition $C_E = 0.5$.

Wilcox *et al.* [4, 5] extended the above analysis in the range of eutectic compositions close to 0.1 and obtained the following results:

$$\frac{\lambda}{\lambda_0} = 1 + A'\varepsilon_0^2 \qquad C_E = 0.1\text{–}0.5$$

where $\varepsilon_0 = G_u\lambda_0^2/D$, λ_0 eutectic spacing without convection and A' a constant.

On the other hand, Quenisset *et al.* [6] studied the directional solidification of Pb–Sn and Cd–Zn eutectics under the existence of convective flow. They found that the eutectic spacing increased with increasing flow velocity. Meanwhile, Jin *et al.* [7] investigated the solidification of Al–Cu eutectic with electromagnetic stirring of the melt. Their experimental results indicated that the eutectic interphase spacing correlated with the flow Peclet number, that is, λ increases if Pe > 1 whereas λ changes little if Pe < 1.

The purpose of the present paper is to obtain an approximate solution which represents the effect of convective flow on eutectic growth under the condition of weak convection. The scheme of the paper is as follows. In Section 2 we establish basic ideas and give the mathematical treatment of this subject. The findings are discussed in Section 3. Then we apply our results to some practical cases in Section 4.

2. MATHEMATICAL TREATMENTS

As mentioned above, several approximate methods have been applied to solve the convection–diffusion equation. Since the equation is nonlinear and complicated, it is too difficult to find an exact analytical solution. However, under some conditions such as weak convection, an approximate solution can be obtained for certain boundary conditions. The so-called weak convection is defined here as the condition for which the flow velocity and its gradient just in front of the S/L interface are both relatively small. There are two types of weak convection in directional solidification: one is weak forced convection caused by weak external mechanical stirring or disk rotating while the other is natural convection, i.e. solutal or thermal convection induced by the difference of solute concentration or temperature during solidification processing.

Generally, any fluid flow parallel to the interface can be resolved into two directions: perpendicular to the lamellae and parallel to the lamellae. Since fluid flow parallel to the lamellae has no effect on solute distribution in the melt if the lamellae are infinite in the same orientation, we consider only the fluid flow perpendicular to the lamellae with a constant gradient of velocity $G_u = \partial u/\partial y$. Furthermore, an assumption suggested by Quenisset *et al.* [2] and Baskaran *et al.* [3] is used that the transverse melt velocity u of interest is presumed to increase linearly with distance from the interface. This is reasonable because in a eutectic melt the composition variation occurs only within a distance of the order of λ, which is very small, typically a few μm. Another assumption is that the S/L interface is planar throughout the solidification processing. Under the condition of weak convection, this allows the complicated convection–diffusion equation, which governs the directional solidification of eutectic alloys, to be transformed into the following non-dimensionalized form with regard to a co-ordinate system (xoy) moving with the S/L interface as shown in Fig. 1 [3]:

$$\frac{\partial^2 C}{\partial X^2} + \frac{\partial^2 C}{\partial Y^2} + \Lambda \frac{\partial C}{\partial Y} - \varepsilon Y \frac{\partial C}{\partial X} = 0 \qquad (1)$$

where C = solute concentration in the melt, $X = x/\lambda$ dimensionless x coordinate, $Y = y/\lambda$ dimensionless y coordinate, $\Lambda = V\lambda/D$ dimensionless interlamellar spacing based Peclet number, and $\varepsilon = G_u\lambda^2/D$ dimensionless convective velocity which is a measure of the intensity of convection.

Since ε is very small under the assumption of weak convection, we can find an approximate solution of the above equation by an asymptotic expansion method [8] using power series as basis functions. Firstly, we propose that the general solution of equation (1) can be described as

$$C(X, Y) = C_0(X, Y) + \sum_{k=1}^{\infty} \varepsilon^k C_k(X, Y) \qquad (2)$$

Then, inserting equation (2) into equation (1) for recombination of terms and setting the coefficient of each power of ε equal to zero, we can obtain

$$\frac{\partial^2 C_0(X, Y)}{\partial X^2} + \frac{\partial^2 C_0(X, Y)}{\partial Y^2} + \Lambda \frac{\partial C_0(X, Y)}{\partial Y} = 0 \qquad (3a)$$

$$\frac{\partial^2 C_1(X, Y)}{\partial X^2} + \frac{\partial^2 C_1(X, Y)}{\partial Y^2} + \Lambda \frac{\partial C_1(X, Y)}{\partial Y} = Y \frac{\partial C_0(X, Y)}{\partial X} \qquad (3b)$$

$$\frac{\partial^2 C_k(X, Y)}{\partial X^2} + \frac{\partial^2 C_k(X, Y)}{\partial Y^2} + \Lambda \frac{\partial C_k(X, Y)}{\partial Y} = Y \frac{\partial C_{k-1}(X, Y)}{\partial X} \quad k = 2, \ldots, \infty \qquad (3c)$$

Generally, in order to obtain the solution $C(X, Y)$, we must solve every order equation of equations (3a)–(c). However, since ε is small, the superposition of zero-order and first-order solutions of equations (3a)–(c) should be precise enough to describe the exact solute distribution in the melt.

The general solution of equation (3a), $C_0(X, Y)$, can be obtained from *Jackson and Hunt*'s analysis (*JH* model) [9] as follows:

$$C_0(X, Y) = C' + B_0' e^{-\Lambda Y} + \sum_{n=1}^{\infty} B_n' \cos(2n\pi X) \exp\left(\left[-\frac{V\lambda}{2D} - \sqrt{\left(\frac{V\lambda}{2D}\right)^2 + (2n\pi)^2}\right] Y\right) \qquad (4)$$

where C', B_0' and B_n' are coefficients of eigenfunction terms to be determined by the boundary conditions. Since $(V/2D) \ll (2n\pi/\lambda)$ is satisfied in

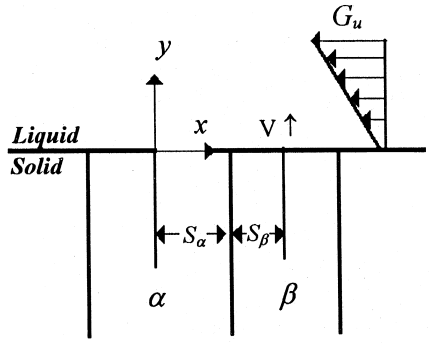

Fig. 1. Schematic illustration of eutectic growth with fluid flow parallel to interface.

practical conditions of solidification processing, equation (4) can be simplified to the following form

$$C_0(X,\ Y) = C' + B_0{}'\mathrm{e}^{-\Lambda Y} + \sum_{n=1}^{\infty} B_n{}' \cos(2n\pi X)\cdot\exp(-2n\pi Y) \quad (5)$$

Inserting equation (5) into equation (3b) and solving the non-homogeneous differential equation as described in Appendix A, we obtain:

$$C_1(X,\ Y) = C_0(X,\ Y) + \sum_{n=1}^{\infty} B_n{}'\frac{1}{\Lambda} \times \left(Y - \frac{2}{\Lambda}\right)\sin(2n\pi X)\cdot\exp(-2n\pi Y) \quad (6)$$

Putting equations (5) and (6) into equation (2) yields the asymptotic expansion solution of equation (1)

$$C(X,\ Y) = C' + (1+\varepsilon)B_0{}'\mathrm{e}^{-\Lambda Y} + \sum_{n=1}^{\infty}(1+\varepsilon)B_n{}'\mathrm{e}^{-2n\pi Y}\left[\cos(2n\pi X) + \frac{\varepsilon}{1+\varepsilon}\frac{1}{\Lambda}\left(Y - \frac{2}{\Lambda}\right)\sin(2n\pi X)\right] \quad (7a)$$

Considering the far field boundary condition, i.e. $C(X,\ Y) = C_\mathrm{E} + C_\infty$ at $Y \to \infty$, we can have $C' = C_\mathrm{E} + C_\infty$ where C_∞ is the difference in composition between initial alloy and eutectic. Furthermore, under the condition of weak convection, i.e. $\varepsilon \ll 1$, there exists $\varepsilon/(1+\varepsilon) \approx \varepsilon$. Thus, equation (7a) can be simplified and written as

$$C(X,\ Y) = C_\mathrm{E} + C_\infty + B_0\mathrm{e}^{-\Lambda Y} + \sum_{n=1}^{\infty} B_n\mathrm{e}^{-2n\pi Y}\left[\cos(2n\pi X) + \frac{\varepsilon}{\Lambda}\left(Y - \frac{2}{\Lambda}\right)\sin(2n\pi X)\right] \quad (7b)$$

where $B_0 = (1+\varepsilon)B_0{}'$, $B_n = (1+\varepsilon)B_n{}'$.

Eutectic growth requires a flux balance in front of the S/L interface, that is,

$$\left.\frac{\partial C}{\partial Y}\right|_{Y=0} = -\Lambda C_0^{\alpha}, \qquad -\frac{S_\alpha}{\lambda} < X < \frac{S_\alpha}{\lambda} \quad (8a)$$

$$\left.\frac{\partial C}{\partial Y}\right|_{Y=0} = \Lambda C_0^{\beta}, \qquad -\frac{1}{2} < X < -\frac{S_\alpha}{\lambda} \text{ and } \frac{S_\alpha}{\lambda} < X < \frac{1}{2} \quad (8b)$$

where S_α and S_β are the half widths of α and β lamellae, respectively, C_0, $C_0{}^\alpha$ and $C_0{}^\beta$ are characteristic concentration ranges which are defined in Fig. 2. These boundary condition equations are integrated between limits $-\frac{1}{2}$ and $\frac{1}{2}$ for X in order to obtain the expressions for B_n in equation (7b). This process is described in Appendix B where the orthogonality of trigonometric functions is used to simplify the mathematical treatments. From this, we obtain:

$$B_0 = \frac{C_0^{\alpha}S_\alpha - C_0^{\beta}S_\beta}{S_\alpha + S_\beta} \quad (9a)$$

$$B_n = \frac{2}{(n\pi)^2}(S_\alpha + S_\beta)\frac{V}{D}C_0\sin\left(\frac{n\pi S_\alpha}{S_\alpha + S_\beta}\right) \quad (9b)$$

From equations (7a)–(b) and (9a)–(b), we can get the solute concentration of the liquid near the interface as,

$$C(X,\ 0) = C_\mathrm{E} + C_\infty + B_0 + \sum_{n=1}^{\infty} B_n\left[\cos(2n\pi X) - \frac{2\varepsilon}{\Lambda^2}\sin(2n\pi X)\right] \quad (10)$$

and the average solute concentration of α and β phases are obtained, respectively, as follows:

$$\overline{C_\alpha} = \frac{1}{S_\alpha}\int_0^{S_\alpha} C(X,\ 0)\,\mathrm{d}x = C_E + C_\infty + B_0 + \frac{2(S_\alpha + S_\beta)^2}{S_\alpha}\frac{V}{D}C_0\left(P - \frac{2\varepsilon}{\Lambda^2}P_\alpha\right) \quad (11a)$$

and

$$\overline{C_\beta} = \frac{1}{S_\beta}\int_0^{S_\beta} C(X,\ 0)\,\mathrm{d}x = C_E + C_\infty + B_0 - \frac{2(S_\alpha + S_\beta)^2}{S_\beta}\frac{V}{D}C_0\left(P - \frac{2\varepsilon}{\Lambda^2}P_\beta\right) \quad (11b)$$

where

$$P = \sum_{n=1}^{\infty}\left(\frac{1}{n\pi}\right)^3\sin^2\left(\frac{n\pi S_\alpha}{S_\alpha + S_\beta}\right) \quad (12a)$$

$$P_\alpha = \sum_{n=1}^{\infty}\left(\frac{1}{n\pi}\right)^3\sin\left(\frac{n\pi S_\alpha}{S_\alpha + S_\beta}\right)\left[1 - \cos\left(\frac{n\pi S_\alpha}{S_\alpha + S_\beta}\right)\right] \quad (12b)$$

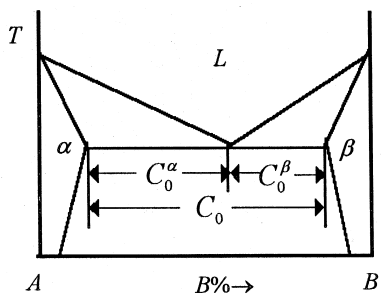

Fig. 2. Phase diagram showing the definitions of C_0, $C_0{}^\alpha$ and $C_0{}^\beta$.

Table 1. Numerical calculations of P, P_α and P_β

$f_\alpha = S_\alpha/(S_\alpha + S_\beta)$	$P = \Sigma_{n=1}^{\infty}(1/n\pi)^3\sin^2(n\pi f_\alpha)$	$P_\alpha = \Sigma_{n=1}^{\infty}(1/n\pi)^3\sin(n\pi f_\alpha)[1-\cos(n\pi f_\alpha)]$	$P_\beta = \Sigma_{n=1}^{\infty}(1/n\pi)^3\sin(n\pi f_\alpha)[(-1)^n - \cos(n\pi f_\alpha)]$
0	0	0	0
0.1	0.00626	0.00626	0.00626
0.2	0.01633	0.01633	0.01633
0.3	0.02552	0.02552	0.02552
0.4	0.03174	0.03174	0.03174
0.5	0.03392	0.03392	0.03392

and

$$P_\beta = \sum_{n=1}^{\infty}\left(\frac{1}{n\pi}\right)^3 \sin\left(\frac{n\pi S_\alpha}{S_\alpha + S_\beta}\right)\left[(-1)^n - \cos\left(\frac{n\pi S_\alpha}{S_\alpha + S_\beta}\right)\right] \tag{12c}$$

Table 1 shows the results of calculations of P, P_α and P_β according to equations (12a)–(c), respectively, indicating that $P = P_\alpha = P_\beta$ for the volume fractions considered, where a small error of 10^{-6} is adopted to identify the convergence of the calculated series.

Using the method given by Jackson and Hunt [9], we can write down the average undercooling of each phase at the interface, neglecting the kinetic undercooling:

$$\Delta T_\alpha = m_\alpha\left[C_\infty + B_0 + \frac{2(S_\alpha + S_\beta)^2}{S_\alpha}\frac{V}{D} \times C_0 P\left(1 - \frac{2\varepsilon}{\Lambda^2}\right)\right] + \frac{a_\alpha^{\mathrm{L}}}{S_\alpha} \tag{13a}$$

$$\Delta T_\beta = m_\beta\left[-C_\infty - B_0 + \frac{2(S_\alpha + S_\beta)^2}{S_\beta}\frac{V}{D} \times C_0 P\left(1 - \frac{2\varepsilon}{\Lambda^2}\right)\right] + \frac{a_\beta^{\mathrm{L}}}{S_\beta} \tag{13b}$$

where m_α and m_β are liquidus slopes of α and β phases, and $a_\alpha{}^{\mathrm{L}}$ and $a_\beta{}^{\mathrm{L}}$ are coefficients of interface tension of α and β phases, respectively.

Letting $\Delta T = \Delta T_\alpha = \Delta T_\beta$, we obtain

$$\frac{\Delta T}{m} = V\lambda Q^{\mathrm{L}} + \frac{a^{\mathrm{L}}}{\lambda} \tag{14}$$

which is similar to the form given by the JH model [9] under no convection, and where

$$\frac{1}{m} = \frac{1}{m_\alpha} + \frac{1}{m_\beta} \tag{15a}$$

$$Q^{\mathrm{L}} = \frac{P(1+\zeta)^2 C_0}{\zeta D}\left(1 - \frac{2\varepsilon}{\Lambda^2}\right) \tag{15b}$$

and

$$a^{\mathrm{L}} = 2(1+\zeta)\left(\frac{a_\alpha^{\mathrm{L}}}{m_\alpha} + \frac{a_\beta^{\mathrm{L}}}{\zeta m_\beta}\right) \tag{15c}$$

The minimum undercooling criterion proposed by Zener [10] here gives

$$\lambda^2 V = \frac{a^{\mathrm{L}}}{Q^{\mathrm{L}}} \tag{16}$$

From the JH model [9] we obtain

$$\lambda_0^2 V = \frac{a^{\mathrm{L}}}{Q_{\mathrm{JH}}^{\mathrm{L}}} \tag{17}$$

where

$$Q_{\mathrm{JH}}^{\mathrm{L}} = \frac{P(1+\zeta)^2 C_0}{\zeta D} \tag{18}$$

Furthermore, comparing equation (16) with equation (17), we obtain

$$\left(\frac{\lambda}{\lambda_0}\right)^2 = \frac{1}{1 - (2\varepsilon/\Lambda^2)} \tag{19a}$$

that is

$$\left(\frac{\lambda}{\lambda_0}\right)^2 = \frac{1}{1 - (2DG_u/V^2)} \tag{19b}$$

Thus, equations (19a)–(b) show the final result of our analysis which predicts the effect of weak convection on interlamellar spacing for eutectic growth.

3. DISCUSSION

3.1. Effect of weak convection on eutectic growth

From the above analysis, it is possible to describe the effect of weak convection on eutectic growth:

(i) Convective flow results in a small shift of solute concentration profile in the liquid near the interface along the direction of fluid flow as shown in Fig. 3, which indicates that the solute field ahead of the interface is not symmetrical to the central surface of the lamellae any longer. Figure 3 is plotted according to the Pb–Sn eutectic system with the parameters listed in Table 2.

(ii) Weak convection tends to decrease the extremum supercooling of eutectic growth, and to increase lamellar spacing, as shown schematically in Fig. 4. Under the same growth velocity, the higher intensity convective flow will induce a more obvious

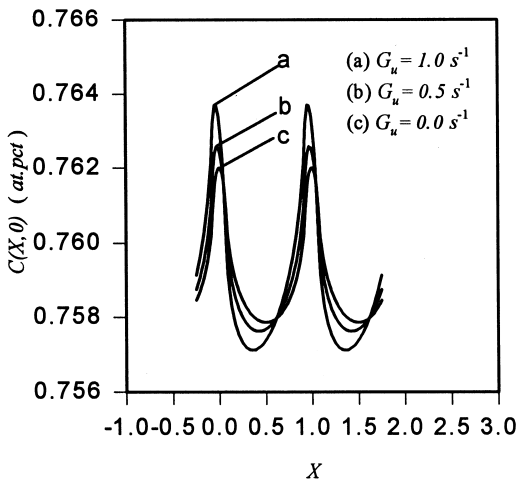

Fig. 3. Effect of weak convection on interface solute concentration of Pb–Sn eutectic at $V = 50\ \mu m \cdot s^{-1}$.

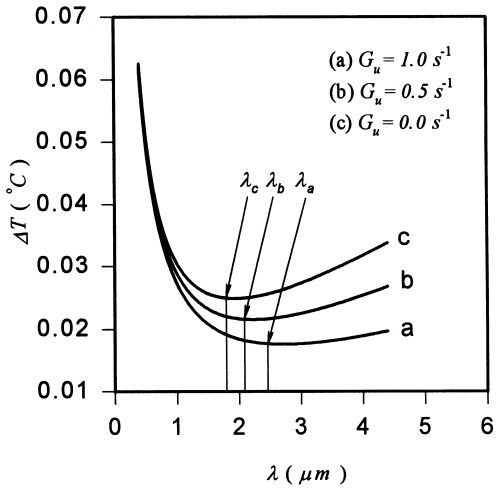

Fig. 4. Effect of weak convection on the growth undercooling of Pb–Sn eutectic growth at $50\ \mu m \cdot s^{-1}$.

decrease in undercooling and increase of interlamellar spacing.

(iii) Figure 5 is drawn from equations (19a)–(b) to show schematically the effect of weak convection on interlamellar spacing of the Pb–Sn eutectic system. A natural variable of equations (19a)–(b) is the dimensionless quantity λ/λ_0, which is plotted vs G_u for several values of growth velocity in Fig. 5(a), and also vs V for various values of the velocity gradient of fluid flow in Fig. 5(b). It can been seen clearly from these two figures that convective flow will result in an increase of interlamellar spacing. Under the same growth velocity, higher intensity convective flow leads to a larger increase in lamellar spacing, but if the flow is too weak, it will have no influence on lamellar spacing. Under the same condition of convection, the lamellar spacing increases more rapidly when the growth velocity is close to the upper limit of weak convection and remains unchanged for higher growth velocity.

3.2. Comparison with early models

Two typical early models on the effect of convection on eutectic growth were proposed by Quenisset *et al.* [3] in 1981 and Wilcox *et al.* [4] in 1984. There are some quantitative differences between their models and ours which we will show as follows.

Quenisset *et al.* obtained

$$\lambda^2 V = \frac{A}{1 - BG_u\lambda^2/D} \tag{20}$$

where $A = 33\ \mu m^3 \cdot s^{-1}$ and $6 \times 10^{-3} < B < 1.5 \times 10^{-2}$ for Pb–Sn eutectic system [6]. With some algebraic transformation in terms of $\lambda_0^2 V = A$, equation (20) can be rewritten as

$$\left(\frac{\lambda}{\lambda_0}\right)^2 = \frac{1 - \sqrt{1 - 4\kappa}}{2\kappa} \tag{21}$$

where $\kappa = ABG_u/VD = (3 \times 10^{-4}\text{–}7.5 \times 10^{-4})G_u/V$ which is in the order of less than 10^{-3} for the order

Table 2. Parameters used in calculations for two typical eutectic systems

Parameter		Pb–Sn eutectic	Al–Cu eutectic
Eutectic composition	C_E	73.90 at.% [11]	17 at.% [11]
Solute diffusion coefficient	D	620 $\mu m^2 \cdot s^{-1}$ [12]	2800 $\mu m^2 \cdot s^{-1}$ [13]
Liquidus slope	m_α	3.95 k/at.% [12, 14]	6.48 k/at.% [15]
	m_β	1.36 k/at.% [12, 14]	6.79 k/at.% [15]
Gibbs–Thomson coefficient	Γ_α	0.0475 k·μm [12, 16]	0.240 k·μm [13]
	Γ_β	0.0785 k·μm [12, 16]	0.055 k·μm [13]
Contact angle	θ_α	35° [13]	65° [13]
	θ_β	65° [13]	55° [13]
Solubility limit	C_0^α	44.90 at.% [11]	14.62 at.% [11]
	C_0^β	14.70 at.% [11]	14.80 at.% [11]
Volume fraction	f_α	0.28 [11, 12]	0.454 [17]
	f_β	0.72 [11, 12]	0.546 [17]
Kinematic viscosity	ν	2.5×10^{-3} $cm^2 \cdot s^{-1}$ [6]	4.885×10^{-3} $cm^2 \cdot s^{-1}$ [18]
Thermal diffusion coefficient	a		0.33 $cm^2 \cdot s^{-1}$ [19]
Prandt number	$Pr = \nu/a$		0.015 [20]
Schmidt number	$Sc = \nu/D$		153 [20]

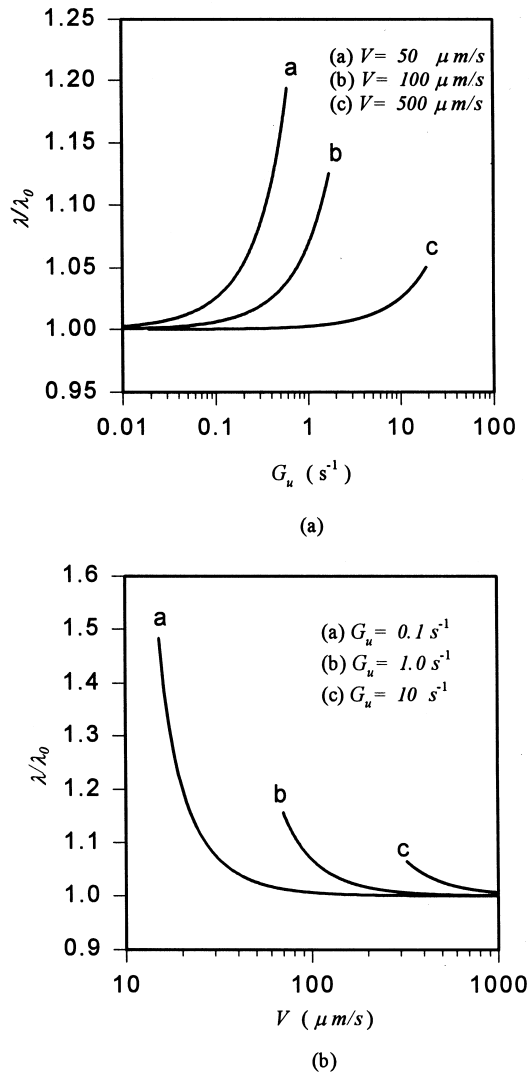

Fig. 5. Effect of weak convection on lamellar spacing of Pb–Sn eutectic: (a) variation of lamellar spacing with weak convection for three growth velocities, (b) variation of lamellar spacing with growth velocity under various weak convections.

of magnitude of G_u and V under weak convection. For such a small value of κ, $\lambda/\lambda_0 \simeq 1$ using equation (21). Therefore, we can conclude that Quenisset *et al.*'s model predicts that eutectic lamellar spacing remains unchanged for the case of weak convection at least for the Pb–Sn system. One probable reason is that the presumed characteristic flow distance in their model is not applicable for the case of weak convection.

Wilcox *et al.* obtained [4]

$$\frac{\lambda}{\lambda_0} = 1 + 0.00034\varepsilon_0^2 \quad \text{for } C_E = 0.5 \tag{22a}$$

and

$$\frac{\lambda}{\lambda_0} = 1 + 0.00028\varepsilon_0^2 \quad \text{for } C_E = 0.333 \tag{22b}$$

where $\varepsilon_0 = G_u\lambda_0^2/D$. Since ε_0 is relatively small under weak convection, usually about 0.1, their model also predicts that weak convective flow will produce little effect on eutectic growth. One probable reason is that their assumption of $\Delta = \Lambda f(\varepsilon)$ may not be suitable for the case of weak convection. Another reason is that the superposed boundary condition at triple phase points, i.e. $\partial C/\partial y = \Lambda(C_i - 0.5)$ is believed not to be applicable because it leads to a discontinuous profile of solute concentration at the interface.

In addition to the above mentioned models, a stagnant film model was first employed in 1970 [2], in which it was assumed that a layer of unstirred melt remains ahead of the interface while beyond this the melt is completely mixed. It predicted that lamellar spacing would increase if the film thickness was less than the lamellar spacing while it would be unchanged if the film thickness was larger than the lamellar spacing. In the next section, we will compare the present model with the stagnant film model with regard to some practical cases of weak convection.

4. APPLICATIONS

Since weak convection is defined in the present analysis as G_u in the order of 10^{-3}–$10\ s^{-1}$, it can be satisfied in directional solidification performed in a horizontal ampoule or by a means of rotating disk. In the former case, natural convection can be produced ahead of the interface, which includes solutal and thermal convection induced by differences of solute concentration and temperature in the melt. This method is especially applicable to the growth of transparent organic materials on a Jackson-type gradient plate. In the latter case, weak forced convection will be induced by rotating the sample uniformly or spin-up (or spin-down) (called rotating disk and the accelerated crucible rotation technique, respectively) if the rotational speed is very low. Therefore, it is of interest and importance to show what effects typical convective situations in practice have on lamellar spacing, and then, conduct some comparisons with the "stagnant film" model.

4.1. Rotating disk

In this case a rotating disk is used as the solidification surface, where weak convection can be produced ahead of the interface by rotating the disk very slowly. The theoretical analysis of convective flow for an infinite disk rotating in a semi-infinite body of fluid [21] gives:

$$G_u = 0.616 r\omega^{3/2}/\nu^{1/2} \tag{23}$$

while the stagnant film thickness for a liquid is:

$$\delta = 1.61(D/\nu)^{1/3}(\nu/\omega)^{1/2} \tag{24}$$

Table 3. Influence of weak convection generated by a rotating disk on lamellar spacing of Al–Cu eutectic where $V = 100\ \mu\text{m}\cdot\text{s}^{-1}$

Rotation rate (rpm)	Radial position (mm)	G_u (s^{-1})	δ (μm)	λ/λ_0 present model	λ/λ_0 stagnant film
5	0.2	0.066	1573	1.019	1.0
5	0.4	0.132	1573	1.039	1.0
5	0.6	0.198	1573	1.060	1.0
5	0.8	0.264	1573	1.083	1.0
5	1.0	0.330	1573	1.108	1.0
10	0.2	0.187	1112	1.056	1.0
10	0.4	0.373	1112	1.124	1.0
10	0.6	0.560	1112	1.207	1.0

Table 4. Influence of natural convection from a heated vertical plate on lamellar spacing of Al–Cu eutectic where $V = 100\ \mu\text{m}\cdot\text{s}^{-1}$

Distance from bottom (mm)	Gr	G_u (s^{-1})	δ (μm)	λ/λ_0 present model	λ/λ_0 stagnant film
10	50	0.504	1186	1.180	1.0
10	100	0.848	998	1.380	1.0
20	50	0.126	2373	1.037	1.0
20	100	0.212	1996	1.065	1.0
20	150	0.287	1803	1.092	1.0
20	200	0.356	1678	1.118	1.0
20	250	0.421	1587	1.144	1.0

where r is radial position, ω is rotation rate, and ν is kinematic viscosity.

Calculated values of λ/λ_0 for typical growth conditions in Al–Cu eutectic system are shown in Table 3. It can be seen that the present model predicts that λ increases from center to endwall along the radial direction, which shows a qualitative agreement with some experiments [6, 7, 22], while the stagnant film model shows both no radial dependence and no influence on lamellar spacing under such weak convection conditions.

4.2. Natural convection

Natural convection is a common phenomenon during solidification processing both on the ground and in microgravity. For simplicity of calculation, we here consider the case of a heated vertical flat plate in a semi-infinite body of liquid [21, 23, 24] for which

$$G_u = \frac{1.92\nu \mathrm{Gr}^{3/4}}{x_0^2 \mathrm{Pr}^{1/4}} \tag{25}$$

and

$$\delta = \frac{x_0 b}{0.667[\mathrm{Pr}^2\mathrm{Gr}/(0.952 + \mathrm{Pr})]^{1/4}} \tag{26}$$

where Gr = Grashof number, Pr = Prandt number, x_0 = distance from bottom, b = a function of Pr/Sc where Sc = Schmidt number.

Calculated values of λ/λ_0 for typical growth conditions of the Al–Cu eutectic system are given in Table 4. For a certain position from the bottom of the plate, the present model shows an increase of lamellar spacing with increase of Gr while the stagnant film model exhibits unaltered lamellar spacing under the same conditions.

5. CONCLUSIONS

An approximate analytical solution has been obtained for the convection–diffusion equation of eutectic growth under the condition of weak convection. The analysis shows that weak convection will increase eutectic lamellar spacing, especially at lower growth velocities, due to the fact that convective flow results in a slight shift of solute concentration field and leads to a decrease of the eutectic growth undercooling. We applied our results to some practical cases such as rotating disk and natural convection to obtain some new results. The results reported here consider only an extremum case of coupling between fluid flow and solute diffusion. To reveal the general effect of convective flow on characteristic solidification lengths such as eutectic lamellar spacing, dendritic primary arm spacing and secondary arm spacing, further theoretical analyses and experimental studies will be performed and reported later.

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APPENDIX A

It is noted that equation (3b) is a non-homogeneous partial differential equation which can be presented as the sum of a particular solution and a solution of the "associated homogeneous equation" which has the same form as equation (3a). Accordingly, $C_1(X, Y)$ can be written as

$$C_1(X, Y) = C_0(X, Y) + C_1^*(X, Y) \tag{A.1}$$

where $C_0(X, Y)$ is the solution of the associated homogeneous equation, and $C_1^*(X, Y)$ is a particular solution of equation (3b). To ascertain $C_1^*(X, Y)$, we substitute equation (5) into equation (3b) and have

$$\frac{\partial^2 C_1^*(X, Y)}{\partial X^2} + \frac{\partial^2 C_1^*(X, Y)}{\partial Y^2} + \Lambda \frac{\partial C_1^*(X, Y)}{\partial Y} = \sum_{n=1}^{\infty} B_n'(-2n\pi) Y e^{-2n\pi Y} \sin(2n\pi X) \tag{A.2}$$

Then we apply the principle of superposition and resolve equation (A.2) into the following form

$$\frac{\partial^2 C_{1n}^*(X, Y)}{\partial X^2} + \frac{\partial^2 C_{1n}^*(X, Y)}{\partial Y^2} + \Lambda \frac{\partial C_{1n}^*(X, Y)}{\partial Y} = (-2n\pi) Y e^{-2n\pi Y} \sin(2n\pi X) \quad \text{for } n = 1, 2, \ldots \tag{A.3}$$

where $C_{1n}^*(X, Y)$ is the eigensolutions of $C_1^*(X, Y)$. Integrating the above equation, we obtain

$$C_{1n} = \frac{1}{\Lambda}\left(Y + \frac{1}{2n\pi} - \frac{2}{\Lambda}\right) e^{-2n\pi Y} \sin(2n\pi X) \quad \text{for } n = 1, 2, \ldots \tag{A.4}$$

Since $(1/2n\pi) \ll 2/\Lambda$ especially for larger n, equation (A.4) can be simplified to

$$C_{1n}^* = \frac{1}{\Lambda}\left(Y - \frac{2}{\Lambda}\right) e^{-2n\pi Y} \sin(2n\pi X) \quad \text{for } n = 1, 2, \ldots \tag{A.5}$$

Hence, combination of the eigensolutions [equation (A.5)] with respective coefficients yields a particular solution

$$C_1^*(X, Y) = \sum_{n=1}^{\infty} B_n' \frac{1}{\Lambda}\left(Y - \frac{2}{\Lambda}\right) e^{-2n\pi Y} \sin(2n\pi X) \tag{A.6}$$

After all, putting equation (A.6) into equation (A.1), we obtain the solution of equation (3b), $C_1(X, Y)$, as described in equation (6).

APPENDIX B

To find the expressions for B_n in equation (7b), we use the boundary conditions given by equations (8a)–(b). The first derivative of $C(X, Y)$ with respect to Y can be obtained by differentiating equation (7b):

$$\left.\frac{\partial C(X, Y)}{\partial Y}\right|_{Y=0} = -B_0\Lambda + \sum_{n=1}^{\infty} B_n \left\{ (2n\pi)\left[-\cos(2n\pi X) + \frac{2\varepsilon}{\Lambda}\sin(2n\pi X)\right] + \frac{\varepsilon}{\Lambda}\sin(2n\pi X)\right\} \tag{B.1}$$

Multiplying equations (8a)–(b) by $\cos(2k\pi X)$, and integrating between limits $-\frac{1}{2}$ and $\frac{1}{2}$ for X, we obtain

$$\int_{-1/2}^{1/2} \left(\left.\frac{\partial C(X, Y)}{\partial Y}\right|_{Y=0}\right)\cos(2k\pi X)\, dX = \int_{-S_z/\lambda}^{S_z/\lambda} (-\Lambda C_0^{\alpha})\cos(2k\pi X)\, dX + \int_{-1/2}^{-S_z/\lambda} (\Lambda C_0^{\beta})\cos(2k\pi X)\, dX + \int_{S_z/\lambda}^{1/2} (\Lambda C_0^{\beta})\cos(2k\pi X)\, dX \tag{B.2}$$

To simplify equation (B.2), we use the orthogonality of trigonometric functions,

$$2\int_{-1/2}^{1/2} \cos(2k\pi X)\cos(2n\pi X)\, dX = \delta_{kn} = \begin{cases} 0 & \text{for } k \neq n, \\ 1 & \text{for } k = n \end{cases} \tag{B.3a}$$

and

$$2\int_{-1/2}^{1/2} \cos(2k\pi X)\sin(2n\pi X)\, dX = 0 \quad \text{for } k = n \text{ and } k \neq n \tag{B.3b}$$

Finally from equations (B.2) and (B.3a)–(b), we can obtain the expressions for B_n as equations (9a)–(b).